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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations Apr/May-2024

Computer Science Engineering

ADVANCED ALGORITHMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

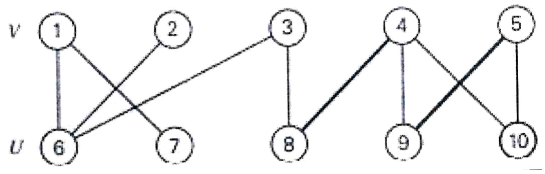
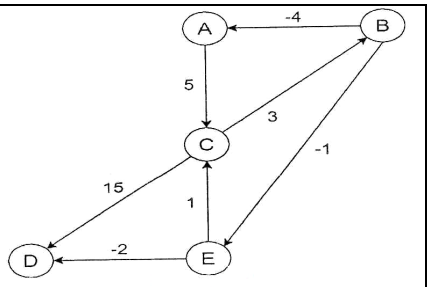
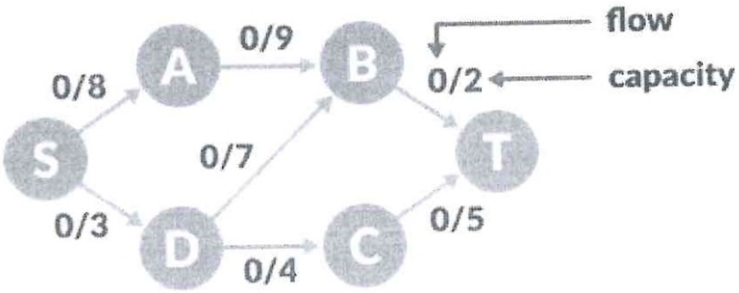
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

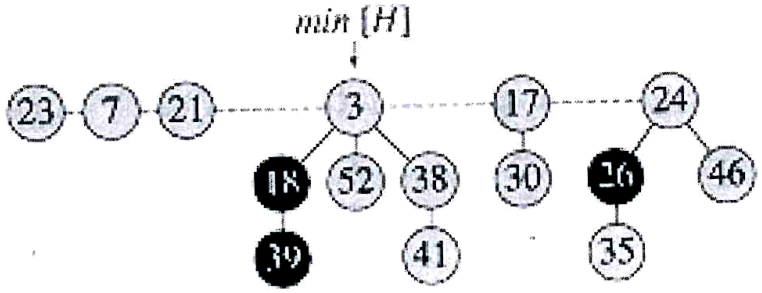
PART-A

			M	BT	CO
1	1.1	Identify the suitable asymptotic notation for the function $f(n) = 4^\wedge(4n)$.	01	1	1
	1.2	Time complexity for the insertion sort is _____.	01	3	1
	1.3	For the recurrence relation $T(1) = 2$, $T(n) = 3T(n/4) + n$ solution for $T(n)$ will be _____.	02	4	2
	1.4	Let $T = 1011101111$ and $P = 111$ find all the Valid Shift using Naïve String matching algorithm. Assume the index of text starts at 0.	02	4	3
	1.5	In a Directed Acyclic Graph (DAG), the total traversal run time is given as _____.	01	1	2
	1.6	For the given $T = 314162349$, $P = 23$, $n = 5$, $m = 2$, $q = 13$, how many spurious hits are generated using Rabin Karp String matching algorithm.	02	4	4
	1.7	Graph 1 has n nodes and it is not complete graph. Graph 2 has m modes and it is a complete graph. The time complexity of Bellman Ford for Graph 1 and Graph 2 are _____ and _____.	02	3	2
	1.8	The run time complexity of Johnson's Algorithm is _____.	01	2	1
	1.9	List any two applications of bipartite graphs.	02	1	3
	1.10	List the two elementary properties of gcd function.	02	1	4
	1.11	The number of trees in a binomial heap with n nodes can be given as _____.	01	2	1
	1.12	The running time for extract-min operation in a Fibonacci heap is _____.	01	1	2
	1.13	The solution for linear congruence $17x \equiv 9\ (mod\ 276)$ is _____.	02	4	1

PART-B

2	a	Draw the recursion tree for $T(n) = 4T(\lfloor n/2 \rfloor) + cn$ where c is a constant and provide a tight asymptotic bound on its solution. Verify the solution by using substitution method.	10	4	1
	b	Explain the accounting method of amortized analysis with the example of a binary counter.	06	2	2
3	a	Write radix sort algorithm and give the analysis. Perform radix sort on the following input: 22 44 33 55 666 777 11 10 12	08	1	1
	b	Consider the input strings "$ABCDGH$" and "$AEDFHR$". Apply LCS algorithm to determine the longest common subsequence using dynamic programming technique. Write the algorithm and explain its complexity.	08	3	2

		OR			
4	a	Write an algorithm to solve matrix chain multiplication using dynamic programming strategy. With the help of an example, give the analysis of the algorithm.	08	4	3
	b	Apply bucket sort on the input array $[0.78, 0.17, 0.39, 0.26, 0.72, 0.94, 0.21, 0.12, 0.23, 0.68]$. Give the analysis of the algorithm.	08	3	3
5	a	For the bipartite graph shown below in Fig. 5.a, find the maximum flow using Ford-Fulkerson algorithm. Identify the flow, capacity, residual capacity, augmented path. Record all the calculations and steps involved.  Fig. 5.a  Fig. 5.b	08	4	3
	b	Apply Johnson's algorithm shown in Fig, 5,b graph. Show the steps.	08	4	3
		OR			
6	a	For the following graph below in Fig. 6.a, apply Ford-Fulkerson algorithm and find the augmented path and the maximum network flow. Draw the residual graph for each path and record all the calculations and steps.  Fig. 6.a	08	3	2
	b	Write and explain the Bellman Ford algorithm and analyze the complexity.	08	2	3
7	a	Apply the Chinese Remainder Theorem to find an x such that: $x \equiv 2(mod\ 5)$ $x \equiv 3(mod\ 7)$ $x \equiv 10(mod\ 11)$	09	3	2
	b	Discuss RSA algorithm in detail.	07	2	1
		OR			
8	a	Apply RSA algorithm for prime numbers 7 and 11 and generate the public and private keys. Show encryption & decryption for $plaintext$ 37.	10	3	4
	b	Write a note on elementary number-theoretic notions.	06	1	1

9	a	Write the naïve string matching algorithm. Show the number of comparisons for the pattern $P = 1001$ in the text $T =$ 00100100001000100100 1. Write the algorithm and give the analysis.	08	3	1
	b	What are the binomial heaps? How do they differ from Fibonacci heaps and binary heaps. Explain with an example.	08	1	1
		OR			
10	a	For working modulo $q = 13$, how many spurious hits does the Rabin-Karp matcher encounter in the text $T = 3141592653589793$ when looking for the pattern $P = 26$? Show the working of the algorithm and give the analysis.	08	4	2
	b	Explain the procedure of extracting next minimum and form the below heap in Fig. 10.b with neat diagrams.  Fig. 10.b	08	2	2